Brent Crude Oil Analysis and Forecasting: Time Series Models and GARCH Volatility, 2006–2025

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Abstract — *This study conducts an in-depth analysis and forecasting of Brent crude oil price volatility from January 2006 to December 2025 to facilitate energy risk management. Utilizing a multivariate monthly time-series dataset comprising 240 observations, the research examines the relationships between oil prices (OIL) and four core macroeconomic variables: the US Dollar Index (USD), Consumer Price Index (CPI), Federal Funds Rate (FED), and Industrial Production Index (IND). The empirical procedure is rigorously implemented across multiple stages, integrating unit root tests, Engle–Granger cointegration tests, structural break tests (CUSUM and Chow tests), and Partial Least Squares (PLS) dimension reduction techniques. The forecasting performance of ARIMA, SARIMA, ARCH, GARCH, GARCH–t, PLS, OLS, and OLS+Dummy models is evaluated using MAE, RMSE, QLIKE, MASE, Theil's* U_2*, and the Diebold–Mariano test. The empirical findings indicate that ARIMA(1,0,1) provides the strongest forecasting accuracy, whereas GARCH-type models more effectively capture volatility clustering and heavy-tailed market behavior. Overall, the study proposes a comprehensive econometric framework for crude oil forecasting and volatility analysis under major global macroeconomic shocks.*

***Keywords* —** *Brent Crude Oil, Oil Price Forecasting, ARIMA, GARCH, Structural Break, Time Series Volatility, Diebold–Mariano Test.*

1. Introduction

1.1. The Importance of Crude Oil Prices in the Global Economy and Historical Shocks During the 2006–2025 Period

Crude oil is not merely a conventional commodity; it has long been regarded as the lifeblood of global energy systems and a fundamental input determining the production cost structure of the world economy. Among international benchmark references, Brent crude oil, traded on the ICE London exchange, plays a dominant pricing role, serving as the benchmark for approximately 60%–70% of globally traded crude oil, particularly for supplies originating from Europe, Africa, and the Middle East. Consequently, fluctuations in Brent crude oil prices generate substantial ripple effects across financial markets and macroeconomic indicators through several important transmission channels.

First, through the input cost channel (supply-side shock), crude oil acts as a direct input for transportation, logistics, petrochemicals, and fertilizer production, while simultaneously functioning as an indirect energy source for industrial activities. As oil prices increase, firms experience higher marginal production costs, shifting the aggregate supply curve leftward and consequently generating cost-push inflation while suppressing economic growth.

Second, through the income and consumption channel (demand-side shock), rising oil prices reduce households' real disposable income in oil-importing economies. This mechanism effectively transfers purchasing power from consuming countries to oil-exporting nations, thereby weakening aggregate demand and slowing economic activity.

Third, through the expectations and monetary policy channel, escalating oil prices intensify inflation expectations and force central banks, particularly the U.S. Federal Reserve, to adopt tighter monetary policies through interest rate increases. These policy adjustments subsequently influence the U.S. dollar index, corporate financing costs, investment decisions, and industrial production, thereby creating a complex and multidimensional network of macroeconomic interactions.

The period from 2006 to 2025 witnessed unprecedented historical disruptions in the global oil market, challenging many traditional linear assumptions regarding oil price dynamics. During the global financial crisis of 2008, Brent crude oil prices collapsed from a historical peak of approximately 147 USD/barrel in July 2008 to below 40 USD/barrel within only a few months. Subsequently, the oil price collapse of 2014–2016 – driven largely by OPEC's decision to maintain high production levels amid the rapid expansion of the U.S. shale revolution – pushed prices from above 100 USD/barrel in mid-2014 to below 30 USD/barrel in early 2016.

The COVID-19 pandemic in 2020 generated the largest demand shock in modern oil market history. For the first time ever, WTI crude oil prices turned negative in April 2020, reaching approximately –37 USD/barrel, while Brent crude oil prices also declined to their lowest level in nearly two decades. Shortly afterward, the Russia–Ukraine conflict in 2022, combined with extensive Western economic sanctions, drove Brent crude oil prices above 130 USD/barrel – the highest level observed since 2008 – before the market entered a prolonged period of restructuring global energy trade patterns that continued through 2025.

These extreme fluctuations, occurring with frequencies substantially higher than predicted under a normal distribution (commonly referred to as "fat tails") and accompanied by strong volatility clustering, present major challenges for conventional oil price forecasting models that assume structural stability over time. Such conditions highlight the necessity of adopting a comprehensive multi-model forecasting framework combined with rigorous statistical testing to improve forecasting accuracy, financial risk management, and macroeconomic policy formulation.

1.2. Research Gap

Although there is a substantial body of research on oil price forecasting, most existing studies focus on one or only a limited number of models, lacking comprehensive comparison and rigorous statistical testing. In particular, few studies simultaneously: (i) test for structural breaks during major crisis events; (ii) integrate PLS dimensionality reduction with macroeconomic variables; (iii) compare linear models, ARIMA/SARIMA, and GARCH volatility models in a unified framework; (iv) apply the Diebold–Mariano test to evaluate forecasting performance differences; and (v) construct Bull/Base/Bear oil price scenarios using bootstrap methods. This study addresses these gaps by implementing a comprehensive 11-stage framework on a 20-year dataset (2006–2025).

1.3. Research Objectives

Given the challenges arising from extreme market volatility and the breakdown of traditional statistical assumptions in crude oil prices during the 2006–2025 period, this study pursues four primary research objectives.

The first objective is to systematically evaluate the statistical properties of the Brent crude oil price series together with relevant macroeconomic variables. This objective is achieved through the application of a comprehensive set of unit root tests, stationarity tests, and distributional diagnostics in order to accurately identify the underlying characteristics of the data prior to model construction.

The second objective focuses on analyzing the volatility structure and the potential existence of structural breaks, particularly during major crisis episodes such as the 2008 global financial crisis, the 2014–2016 oil price collapse, the COVID-19 pandemic in 2020, and the Russia–Ukraine conflict in 2022. Through this analysis, the study aims to quantify the effects of major exogenous shocks on oil price dynamics.

The third objective is to conduct a quantitative comparison of the forecasting performance of eight different econometric models, including both univariate time-series approaches and multivariate frameworks. This comparison is implemented using a rigorous out-of-sample evaluation procedure based on standardized forecasting error metrics together with the Diebold–Mariano statistical significance test.

The fourth and final objective is application-oriented. Specifically, the study seeks to construct oil price forecasting scenarios accompanied by confidence intervals, thereby providing a quantitative foundation for financial risk management in energy-related enterprises as well as for macroeconomic policy formulation by national policymakers, particularly under conditions of increasing geopolitical uncertainty.

2. Literature Review

Empirical literature on crude oil price forecasting can be grouped into three main streams.

2.1. Univariate Time-Series Models

Univariate time-series models rely exclusively on the historical path of oil prices. Hamilton (2009) and Kilian (2009) demonstrated that oil price dynamics are highly nonlinear and dominated by structural shocks, making conventional linear models unreliable during crises. Comparative studies (e.g., Aloui & Jammazi, 2009) showed that regime-switching frameworks (e.g., Markov-switching ARIMA) outperform standard specifications in periods of financial turbulence.

2.2. Volatility Modeling

Volatility modeling focuses on time-varying conditional variance. Following Bollerslev (1986), GARCH-type models and their extensions (EGARCH, GJR-GARCH) have been widely applied to energy markets. Evidence from Kang et al. (2009) and Vo (2009) indicates strong volatility clustering and long-memory persistence in crude oil returns. However, models assuming conditional normality systematically underestimate tail risks, especially during extreme episodes such as the 2020 COVID-19 crash or the 2022 energy crisis. Mohammadi and Su (2010) advocate for fat-tailed distributions (Student-t or GED) to better capture extreme movements.

2.3. Multivariate Models with Macroeconomic Fundamentals

Multivariate models with macroeconomic fundamentals incorporate variables such as interest rates, exchange rates, inflation, and industrial production. Frankel (2006) and Bernanke, Gertler and Watson (1997) highlighted the channels through which monetary policy affects commodity prices. Kilian (2009) decomposed oil price shocks into supply- and demand-side components and showed that real economic activity indicators significantly improve long-horizon forecasts. Nonetheless, the predictive content of US-centric macro variables for monthly oil returns remains contested.

2.4. Summary of Research Gaps

In summary, while each stream has contributed important insights, most existing studies lack a comprehensive pre-estimation diagnostic framework (e.g., joint stationarity and cointegration tests, structural break detection at historically identified crisis points). Furthermore, systematic comparisons across model classes—univariate time-series, volatility, and macro-augmented regressions—are rarely accompanied by rigorous statistical tests such as the Diebold–Mariano procedure. The construction of scenario-based price paths with bootstrap confidence intervals for risk management purposes is also underdeveloped. The present study directly addresses these methodological gaps.

3. Data and Research Methodology

3.1. Data Description and Sources

3.1.1. Variable Description and FRED Codes

This study investigates the relationship between Brent crude oil prices and several major macroeconomic indicators. Five variables are employed in the empirical analysis, including Brent crude oil prices (OIL), the U.S. Dollar Index (USD), the Consumer Price Index (CPI), the Federal Funds Rate (FED), and the Industrial Production Index (IND). All variables are collected from the Federal Reserve Economic Data (FRED) database maintained by the Federal Reserve Bank of St. Louis.

Table 1. Description of Variables and Data Sources

Variable	Description / Source	Unit	Role
OIL	Brent Crude Oil Price (DCOILBRENTEU – FRED)	USD/barrel	Target variable
USD	Trade Weighted U.S. Dollar Index (DTWEXBGS – FRED)	Index	Explanatory variable
CPI	U.S. Consumer Price Index (CPIAUCSL – FRED)	Index	Explanatory variable
FED	Federal Funds Rate (FEDFUNDS – FRED)	% per annum	Explanatory variable
IND	Industrial Production Index (INDPRO – FRED)	Index	Explanatory variable

3.1.2. Sample Period, Frequency, and Observations

This study employs monthly data spanning from January 2006 to December 2025, resulting in a total of 240 observations after data cleaning and merging procedures. Monthly frequency is selected in order to balance the level of informational detail with the reduction of short-term random noise commonly observed in higher-frequency financial data.

3.1.3. Descriptive Statistics

Table 2 presents the descriptive statistics of the variables in their original level form over the sample period.

Table 2. Descriptive Statistics of Variables in Level Form

Statistic	OIL	CPI	USD	FED	IND
Observations	240	240	240	240	240
Mean	76.96	248.64	106.08	1.736	98.71
Median	74.34	238.03	109.35	0.400	100.07
Standard Deviation	23.88	34.83	12.39	1.991	4.206
Coefficient of Variation (%)	31.03	14.01	11.68	114.70	4.26
Minimum	14.85	199.30	85.60	0.050	84.56
25th Percentile	60.82	221.01	93.67	0.120	98.03
75th Percentile	93.80	262.21	116.27	3.895	101.33
Maximum	138.40	326.03	129.28	5.330	104.10
Range	123.55	126.73	43.68	5.280	19.54
Skewness	0.218	0.773	-0.037	0.802	-1.604
Kurtosis	-0.575	-0.454	-1.449	-1.025	2.167

The Brent crude oil price (OIL) exhibits a relatively high degree of volatility, with an average value of 76.96 USD/barrel and fluctuations ranging from a minimum of 14.85 USD/barrel to a maximum of 138.40 USD/barrel. The coefficient of variation reaches approximately 31%, indicating substantial relative variability over the sample period.

The Federal Funds Rate (FED) displays extremely high relative volatility, with a coefficient of variation exceeding 114%, reflecting substantial monetary policy cycles over the sample period. In contrast, the Industrial Production Index (IND) demonstrates relatively stable long-run growth, with a very low coefficient of variation of 4.26%.

Overall, all variables in level form exhibit strong trending behavior and are likely to be non-stationary. Consequently, transforming the variables into log-return form becomes necessary before conducting regression and time-series analyses.

3.2. Data Processing Procedure

3.2.1. Reading and Standardizing Raw `.xlsx` Files

Each raw `.xlsx` file is imported individually, followed by column name standardization and date format conversion to ensure consistency across datasets.

3.2.2. Resampling to Month-End Frequency

All variables are resampled to month-end frequency by selecting the final observation of each month. This approach ensures synchronization among variables while preserving economically meaningful monthly information.

3.2.3. Handling Missing Values

Missing observations are handled sequentially using a three-stage procedure: (i) forward-fill interpolation with a maximum limit of three consecutive months, (ii) backward-fill interpolation with the same three-month restriction, and (iii) removal of any remaining missing observations using `dropna`.

3.2.4. Transformation into Log-Returns

All variables are transformed into log-return form according to the following equation:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \tag{1}$$

where P_t represents the value of the variable at time t, and P_{t-1} denotes its value in the previous period.

This transformation is widely used in financial econometrics because it often converts non-stationary series into stationary processes $I(0)$, while also providing an interpretation in terms of continuously compounded returns.

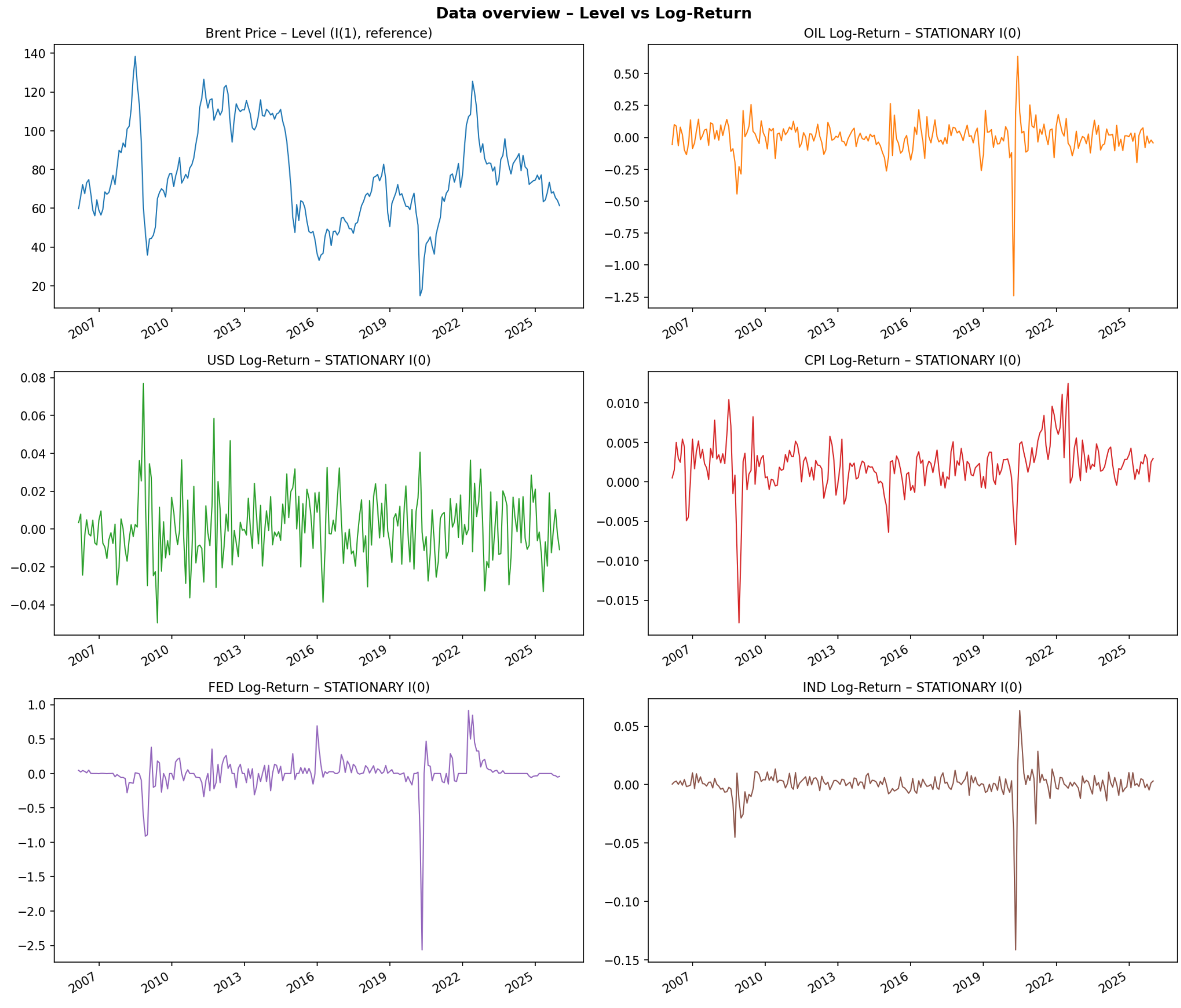

Figure 1. Illustration of the transformation from level data to log-return series

3.3. Theoretical Framework for Variable Selection

3.3.1. U.S. Dollar Index (USD)

The inverse relationship between crude oil prices and the U.S. dollar has been widely documented in the literature. Since Brent crude oil is denominated in U.S. dollars, an appreciation of the dollar increases the effective oil price in foreign currencies, thereby reducing global demand and exerting downward pressure on oil prices.

3.3.2. Consumer Price Index (CPI)

Economic theory identifies crude oil as one of the most important production inputs in the global economy, implying that fluctuations in oil prices are transmitted through production costs into consumer prices. Elevated CPI levels may also increase the attractiveness of commodities as inflation-hedging assets.

3.3.3. Federal Funds Rate (FED)

Interest rates affect crude oil prices through multiple transmission channels, including inventory holding costs, economic growth, aggregate demand, and exchange rate dynamics. Monetary policy therefore plays a crucial role in explaining oil price fluctuations.

3.3.4. Industrial Production Index (IND)

Industrial production serves as a proxy for real economic activity and aggregate oil demand. Demand-side oil market theory identifies industrial production as one of the most important indicators of global energy consumption.

3.4. ADF and KPSS Stationarity Tests

The Augmented Dickey–Fuller (ADF) test is employed to examine the presence of a unit root in each time series:

$$\Delta y_t = \alpha + \beta t + \rho y_{t-1} + \sum_{i=1}^{p} \gamma_i \Delta y_{t-i} + \varepsilon_t \tag{2}$$

The optimal lag length p is selected using the Akaike Information Criterion (AIC). At the 5% significance level, the approximate critical value is -2.86.

The KPSS test adopts the opposite framework, where the null hypothesis assumes stationarity. For the sample size of 240 observations, the 5% critical value is approximately 0.463.

This study combines ADF and KPSS results in order to obtain more robust conclusions regarding the integration properties of the variables.

Table 3. ADF and KPSS Test Results for Level and Log-Return Variables

Form	Variable	ADF Stat	ADF p-value	KPSS Stat	KPSS p-value	Conclusion	I(d)
Level	OIL	-2.4748	0.5000	0.4200	0.1000	Non-stationary	I(1)
	USD	-0.9333	0.5000	3.5103	0.0100	Non-stationary	I(1)
	CPI	2.0980	0.5000	3.7207	0.0100	Non-stationary	I(1)
	FED	-1.8320	0.5000	0.7727	0.0100	Non-stationary	I(1)
	IND	-2.1941	0.5000	0.8994	0.0100	Non-stationary	I(1)
Log-return	OIL_RET	-13.3066	0.0100	0.0369	0.6000	Stationary	I(0)
	USD_RET	-14.5593	0.0100	0.0858	0.6000	Stationary	I(0)
	CPI_RET	-8.4291	0.0100	0.4432	0.1000	Stationary	I(0)
	FED_RET	-10.0860	0.0100	0.2330	0.6000	Stationary	I(0)
	IND_RET	-12.5012	0.0100	0.0445	0.6000	Stationary	I(0)

3.5. Engle–Granger Cointegration Test

The Engle–Granger cointegration procedure is implemented to determine whether long-run equilibrium relationships exist between Brent crude oil prices and the selected macroeconomic variables.

The first step estimates the following OLS regression:

$$OIL_t = \alpha + \beta X_t + \varepsilon_t \tag{3}$$

The second step applies the ADF test to the estimated residual series $\hat{\varepsilon}_t$.

Table 4. Engle–Granger Cointegration Test Results

Pair	EG Statistic	p-value	Lag	Conclusion
OIL ~ USD	-2.8137	0.5000	15	Not cointegrated
OIL ~ CPI	-2.5141	0.5000	15	Not cointegrated
OIL ~ FED	-2.4930	0.5000	15	Not cointegrated
OIL ~ IND	-2.3104	0.5000	15	Not cointegrated

3.6. CUSUM and Chow Structural Break Tests

The CUSUM test evaluates parameter stability based on recursive residuals and a 5% confidence band.

Table 5. CUSUM Stability Test Results for `OIL_RET`

Statistic	Value
Standard deviation of recursive residuals (σ)	0.133216
Observations exceeding 5% confidence boundary	0 / 238
Conclusion	No significant instability

The Chow test statistic is calculated as follows:

$$F = \frac{\left(SSR_p - (SSR_1 + SSR_2)\right)/k}{(SSR_1 + SSR_2)/(n-2k)} \sim F(k, n-2k) \tag{4}$$

Four major historical break points are examined.

Table 6. Chow Structural Break Test Results for `OIL_RET`

Event	Break Date	n_{pre}	n_{post}	F-statistic	p-value	Decision
GFC_2008	2008-09-01	31	208	0.7367	0.391585	No structural break
OilCrash_2014	2014-11-01	105	134	0.0870	0.768233	No structural break
COVID_2020	2020-03-01	169	70	0.0405	0.840758	No structural break
UkraineWar_2022	2022-02-01	192	47	0.2458	0.620501	No structural break

3.7. Jarque–Bera Normality Test

The Jarque–Bera (JB) test is employed to examine whether the log-return series follow a normal distribution. The JB statistic is constructed based on the skewness (S) and kurtosis (K) of the series as follows:

$$\text{JB} = n\left[\frac{S^2}{6} + \frac{(K-3)^2}{24}\right] \sim \chi^2(2) \tag{5}$$

Table 7 reports the empirical Jarque–Bera test results for the five log-return variables.

Table 7. Jarque–Bera Normality Test Results

Variable	JB Statistic	JB p-value	Skewness	Excess Kurtosis	Normality
OIL_RET	11474.3554	0.000000	-3.1606	33.3509	No
USD_RET	28.6694	0.000001	0.4873	1.3890	No
CPI_RET	761.2043	0.000000	-1.1835	8.4164	No
FED_RET	22942.8110	0.000000	-4.5416	47.1315	No
IND_RET	39880.3311	0.000000	-5.3541	62.3703	No

All variables reject the null hypothesis of normality at the 1% significance level ($p < 0.001$). In particular, the OIL_RET series exhibits extremely high excess kurtosis (33.35), indicating pronounced fat-tail behavior. This implies that the probability of extreme oil-price movements is substantially greater than under the Gaussian distribution, especially during extraordinary shocks such as the COVID-19 crisis in April 2020.

These empirical findings suggest that models assuming normally distributed residuals, such as conventional OLS regression, may underestimate tail risk. Consequently, the results provide strong econometric justification for the use of GARCH models with Student-t distributed innovations in the subsequent volatility analysis.

3.8. ARCH–LM Test and ACF/PACF Analysis

3.8.1. ARCH–LM Test

The ARCH–LM test proposed by Engle (1982) is conducted to detect the presence of conditional heteroskedasticity in the OIL_RET series. The auxiliary regression is specified as:

$$\varepsilon_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_p \varepsilon_{t-p}^2 + u_t \tag{6}$$

where the null hypothesis is:

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_p = 0$$

indicating the absence of ARCH effects.

Given the monthly frequency of the data, the lag order is set to 12. The ARCH–LM test applied to the full sample of 239 observations produces a test statistic of 16.32 with a p-value of 0.177, implying insufficient evidence to reject the null hypothesis.

However, after excluding the COVID-19 period (ex-COVID sample), the test statistic increases substantially to 46.79 with $p < 0.001$, leading to rejection of the null hypothesis and confirming the existence of significant ARCH effects in the OIL_RET series.

This result suggests that the COVID-19 shock behaves as an extreme outlier that masks the underlying conditional heteroskedasticity structure of the series. Such a phenomenon, commonly referred to as the masking effect, further supports the appropriateness of employing GARCH-type volatility models.

3.8.2. ACF and PACF Analysis (Yule–Walker)

The autocorrelation function (ACF) and partial autocorrelation function (PACF) are estimated using the Yule–Walker approach in order to identify the appropriate ARMA structure for OIL_RET and to examine volatility clustering through the squared return series OIL_RET^2.

For the OIL_RET series, both the ACF and PACF at lag 1 exceed the approximate 95% significance threshold of:

$$\pm\frac{1.96}{\sqrt{n}} \approx \pm 0.13$$

whereas higher-order lags remain within the confidence interval. This pattern is characteristic of an ARMA(1,1) process, indicating that a single autoregressive and moving-average lag captures most of the serial dependence structure. Therefore, the ARIMA(1,0,1) specification is considered appropriate.

In addition, the Ljung–Box test applied to OIL_RET with 12 lags produces:

$$Q = 14.82, \qquad p = 0.251$$

confirming the absence of statistically significant serial autocorrelation.

For the squared return series OIL_RET^2, the lag-1 autocorrelation is approximately 0.25, substantially exceeding the confidence threshold. This finding provides evidence of volatility clustering, implying that large shocks tend to be followed by additional large shocks.

The Ljung–Box test on OIL_RET^2 yields:

$$Q = 15.47, \qquad p = 0.217$$

Although the p-value is not statistically significant, likely due to the masking effect induced by the COVID-19 outlier, the combined evidence from the ACF structure and the ARCH–LM results after excluding COVID strongly supports the existence of conditional heteroskedasticity and justifies the use of GARCH-type volatility models.

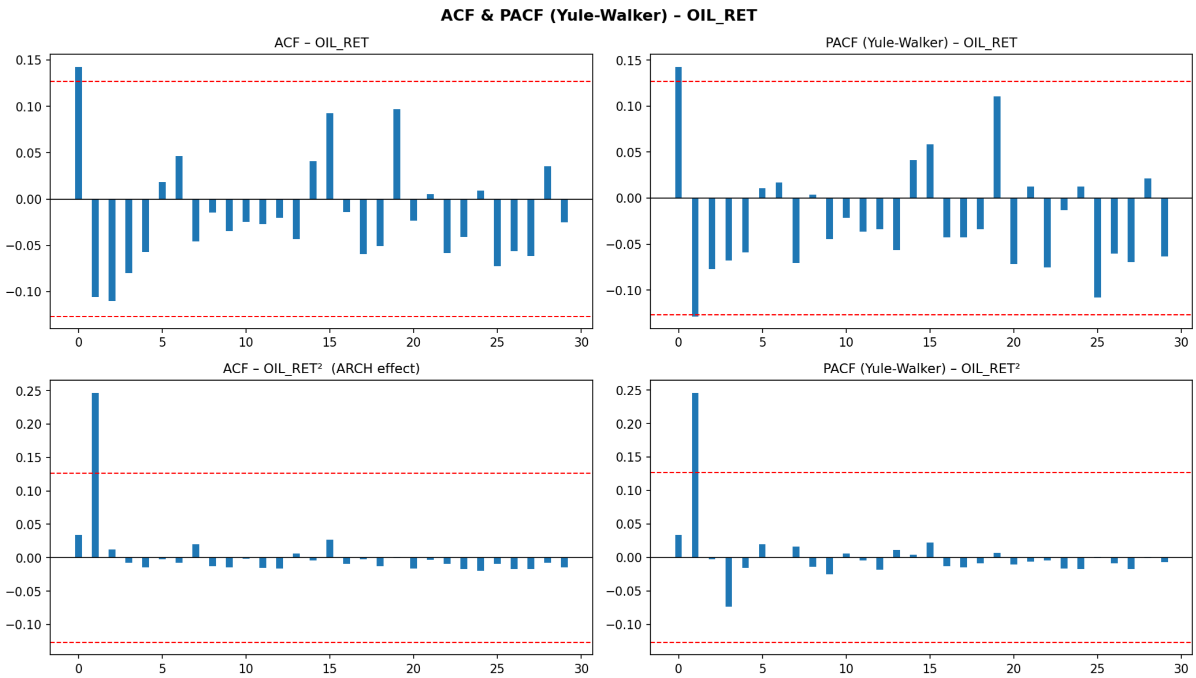

Figure 2. ACF and PACF (Yule–Walker) of OIL_RET and OIL_RET^2

3.9. PLS Dimension Reduction

Partial Least Squares (PLS), originally introduced by Wold (1975), is a supervised dimension-reduction technique that differs fundamentally from Principal Component Analysis (PCA). While PCA focuses solely on maximizing the explained variance of the predictor matrix X, PLS simultaneously: (i) explains the variance of X, and (ii) maximizes the covariance between the extracted components and the target variable Y.

PLS is particularly suitable in the presence of multicollinearity among explanatory variables, which is commonly observed among macroeconomic indicators such as inflation, interest rates, exchange rates, and industrial production.

In this study, the four macroeconomic log-return variables:

[USD_RET, CPI_RET, FED_RET, IND_RET]

form the predictor matrix X of dimension 239×4, while OIL_RET serves as the dependent variable Y.

The PLS procedure sequentially extracts four latent components, where each component is constructed as a linear combination of the explanatory variables that maximizes covariance with the target variable or with the residual variation of the target after removing previous components.

Table 8 presents the variance decomposition associated with the extracted PLS components.

Table 8. Variance Decomposition by PLS Components

PLS Component	Explained Variance	Ratio (%)	Cumulative (%)
PLS1	1.6595	41.31	41.31
PLS2	1.2608	31.39	72.70
PLS3	0.7493	18.65	91.36
PLS4	0.3472	8.64	100.00

The four extracted PLS components jointly explain 100% of the variance in the predictor matrix X. Among them, PLS1 is the most influential component, accounting for 41.31% of the total variance and representing a broad macroeconomic factor related to economic cycles and monetary policy conditions.

The first two components (PLS1 and PLS2) together explain 72.70% of the total variation, capturing the majority of economically relevant information contained in the macroeconomic variables. The remaining components (PLS3 and PLS4) account for the residual variation associated with more idiosyncratic characteristics of the individual variables.

The extracted PLS components are subsequently employed as explanatory variables in the forecasting models developed in the following sections.

4. Forecasting Models and Empirical Results

This section develops and evaluates a comprehensive forecasting framework for Brent crude oil returns. The models are classified into three major groups: (i) macroeconomic regression models using PLS latent factors, (ii) univariate time-series models, and (iii) conditional volatility models.

All models are estimated using the stationary log-return series OIL_RET.

4.1. Crisis Dummy Variables

Based on the major historical events discussed in Chapter I, four dummy variables are constructed to capture the direct impact of major economic and geopolitical crises on Brent crude oil returns.

Each dummy variable takes the value:

$$D_t = \begin{cases} 1, & \text{during the crisis period} \\ 0, & \text{otherwise} \end{cases}$$

The crisis periods are selected according to the duration of the dominant economic effects associated with each event.

Table 9. Crisis Dummy Variables and Estimated Effects in the OLS+Dummy Model

Dummy Variable	Period	Months	Estimated Coefficient	Economic Interpretation
D_GFC	09/2008 – 06/2009	10	+0.01619	The Global Financial Crisis increased average oil returns by approximately 1.6%, reflecting extreme market instability followed by short-term recovery dynamics.
$D\_OIL2014$	11/2014 – 01/2016	15	–0.01616	The 2014–2016 oil supply shock reduced average oil returns by roughly 1.6%, primarily due to prolonged oversupply associated with OPEC production decisions and the expansion of U.S. shale oil production.
D_COVID	03/2020 – 12/2020	10	–0.00320	COVID-19 exerted only a small negative impact on average returns despite the unprecedented collapse in April 2020, suggesting that the crisis mainly affected volatility rather than the conditional mean.
$D\_UKRAINE$	02/2022 – 12/2022	11	≈ 0.00000	The Ukraine war generated offsetting upward and downward price movements, resulting in an economically negligible average effect on monthly oil returns.

The estimated coefficients of D_GFC and $D\_OIL2014$ exhibit nearly symmetric magnitudes with opposite signs, indicating that the two crises exerted comparable but opposite directional impacts on oil returns.

The coefficient associated with D_COVID remains relatively small compared with the magnitude of the actual oil-price collapse observed during April 2020. This finding implies that the COVID-19 shock primarily affected conditional variance rather than average returns.

Similarly, the coefficient of $D\_UKRAINE$ is approximately zero because the strong positive oil-price movements observed immediately after the invasion were subsequently offset by later market corrections.

Overall, the inclusion of crisis dummy variables produces only marginal changes in the regression coefficients and does not materially improve forecasting performance.

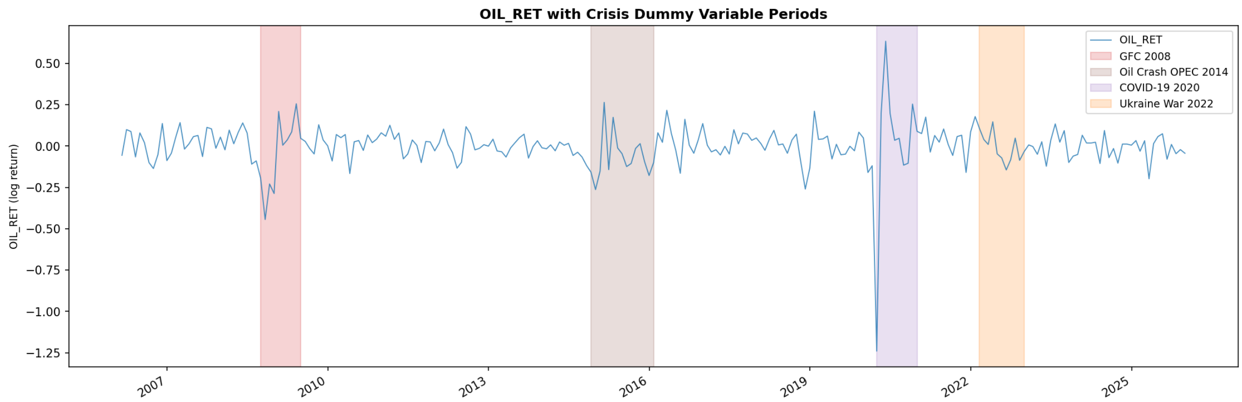

Figure 3. OIL_RET with highlighted crisis dummy periods: GFC 2008, Oil Crisis 2014, COVID-19, and Ukraine War

4.2. Sample Splitting and Walk-Forward Rolling Procedure

The complete log-return dataset contains 239 monthly observations spanning from February 2006 to December 2025.

To evaluate genuine out-of-sample forecasting performance while preventing information leakage from future observations, the dataset is divided into training and testing subsets according to a walk-forward expanding-window procedure.

- **Training set:** February 2006 – December 2020 (179 observations)
- **Testing set:** January 2021 – December 2025 (60 observations)

A walk-forward rolling forecasting strategy with eight rolling windows is implemented.

At the first iteration, the model is estimated using observations from February 2006 to December 2020 and forecasts are generated for the subsequent seven months.

The training window is then expanded forward by seven observations, after which the model is re-estimated and new forecasts are produced for the next seven months.

This process is repeated sequentially until the entire testing sample is exhausted.

The walk-forward approach closely resembles real-world forecasting environments because only information available at each point in time is utilized. Consequently, the procedure eliminates look-ahead bias and allows forecasting stability to be evaluated across different market conditions.

4.3. Forecasting Models

Eight forecasting models are constructed and classified into three broad categories:

- Macroeconomic regression models: OLS, PLS, and OLS+Dummy
- Univariate time-series models: ARIMA and SARIMA
- Conditional volatility models: ARCH and GARCH

Although GARCH models are primarily designed to forecast conditional variance, the mean-equation forecasts are also evaluated using RMSE-based forecasting metrics in order to facilitate comparison with alternative specifications.

4.3.1. OLS and PLS Regression Models

The benchmark OLS specification employs the four extracted PLS latent components as explanatory variables:

$$\mathrm{OIL\_RET}_t = \beta_0 + \beta_1 \mathrm{PLS1}_t + \beta_2 \mathrm{PLS2}_t + \beta_3 \mathrm{PLS3}_t + \beta_4 \mathrm{PLS4}_t + \varepsilon_t \tag{7}$$

The PLS forecasting model uses the same latent components but estimates the regression using the PLS algorithm itself rather than ordinary least squares.

Because the number of extracted components equals the number of original macroeconomic variables, the PLS and OLS specifications become mathematically equivalent in this setting.

Consequently, both models produce identical forecasting performance:

$$RMSE = 0.0938, \qquad U_2 = 0.8542$$

These forecasting errors are substantially larger than those obtained from the univariate time-series models.

The results suggest that U.S. macroeconomic variables, even after dimensionality reduction through PLS, provide limited predictive information for short-run monthly Brent oil returns.

This finding is economically plausible because short-term oil-price movements are heavily influenced by geopolitical shocks, OPEC production decisions, inventory dynamics, and unexpected global events rather than conventional macroeconomic indicators alone.

4.3.2. ARIMA(1,0,1)

Based on the ACF/PACF analysis and Akaike Information Criterion (AIC), the ARIMA(1,0,1) model is selected as the preferred univariate specification:

$$\mathrm{OIL\_RET}_t = c + \phi_1 \mathrm{OIL\_RET}_{t-1} + \theta_1 \varepsilon_{t-1} + \varepsilon_t \tag{8}$$

The estimated coefficients obtained from the training sample are:

$$c = -0.001047$$

$$\phi_1 = -0.2342$$

$$\theta_1 = 0.4190$$

The near-zero constant term is consistent with the long-run expectation that average monthly oil returns fluctuate around zero.

The negative autoregressive coefficient indicates weak mean-reversion behavior, implying that positive returns are often followed by mild corrections in subsequent periods.

Meanwhile, the positive moving-average coefficient suggests short-term shock persistence or momentum effects.

Among all competing models, ARIMA(1,0,1) achieves the best forecasting performance:

$$RMSE = 0.0793, \qquad U_2 = 0.7219$$

These results indicate that historical oil-return dynamics alone contain substantial predictive information for short-horizon monthly forecasting.

4.3.3. SARIMA(1,0,1)(1,0,1,12)

The SARIMA specification extends the ARIMA framework by incorporating annual seasonal dynamics:

$$(1-\phi_1 L)(1-\Phi_1 L^{12})\text{OIL\_RET}_t = c + (1+\theta_1 L)(1+\Theta_1 L^{12})\varepsilon_t \tag{9}$$

The seasonal structure is intended to capture recurring annual oil-demand patterns related to heating demand, refinery maintenance cycles, and transportation seasonality.

The SARIMA model produces:

$$RMSE = 0.0809, \qquad U_2 = 0.7363$$

Although the forecasting accuracy remains competitive, the seasonal coefficients are statistically weak and provide only marginal improvement relative to the non-seasonal ARIMA model.

This result suggests that seasonal effects in Brent crude oil returns are relatively limited compared with the dominant influence of geopolitical and macroeconomic shocks during the 2021–2025 testing period.

4.3.4. ARCH(1) and GARCH(1,1)

Conditional volatility models are estimated to capture time-varying oil-market uncertainty and volatility clustering.

The GARCH(1,1) specification is defined as:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 \tag{10}$$

The estimated parameters are:

$$\mu = 0.0199$$

$$\omega = 0.003994$$

$$\alpha = 0.8739$$

$$\beta = 0.1061$$

The volatility persistence coefficient is:

$$\alpha + \beta = 0.9800$$

which is extremely close to unity, indicating highly persistent volatility dynamics.

The corresponding half-life of a volatility shock is approximately:

$$\frac{\ln(0.5)}{\ln(0.98)} \approx 34 \text{ months}$$

This implies that the effects of major shocks such as COVID-19 remain influential for nearly three years.

The ARCH(1) model captures short-run volatility responses but lacks the lagged conditional variance component required to model long-run persistence.

Forecasting performance is summarized as follows:

$$ARCH(1): \quad RMSE = 0.0811$$

$$GARCH(1,1): \quad RMSE = 0.0810$$

Although these RMSE values are slightly inferior to ARIMA, the principal advantage of GARCH models lies in their ability to estimate conditional variance, which is critically important for risk management, derivative pricing, portfolio allocation, and stress testing.

4.3.5. OLS + Dummy Variables

The extended OLS+Dummy specification incorporates the four crisis dummy variables into the baseline regression framework:

$$\text{OIL\_RET}_t = \beta_0 + \sum_{i=1}^{4} \beta_i \text{PLS}_{i,t} + \gamma_1 D\_GFC + \gamma_2 D\_OIL2014 + \gamma_3 D\_COVID + \gamma_4 D\_UKRAINE + \varepsilon_t \quad (11)$$

The inclusion of dummy variables does not materially improve forecasting performance:

$$RMSE = 0.0940, \qquad U_2 = 0.8558$$

In fact, forecasting accuracy deteriorates slightly relative to the baseline OLS specification.

This result confirms that crisis events primarily influence volatility rather than the conditional mean of oil returns.

Consequently, simply introducing crisis dummies into the mean equation is insufficient for capturing the extreme uncertainty characterizing oil-price dynamics.

4.4. Forecast Performance Comparison

Table 10 summarizes the out-of-sample forecasting performance of all competing models.

Table 10. Forecast Performance Comparison Across Models

Model	RMSE	Theil U_2	Rank
ARIMA	**0.0793**	**0.7219**	1
GARCH-t	0.0796	0.7243	2
SARIMA(1,0,1)(1,0,1,12)	0.0809	0.7363	3
GARCH(1,1)	0.0810	0.7370	4
ARCH(1)	0.0811	0.7378	5
OLS	0.0938	0.8542	6
PLS	0.0938	0.8542	7
OLS+Dummy	0.0940	0.8558	8

Overall, ARIMA delivers the best forecasting performance according to both RMSE and Theil U_2.

However, the difference between ARIMA and GARCH-t is economically small:

$$0.079566 - 0.079307 = 0.000259$$

which corresponds to a negligible improvement in forecasting accuracy.

As demonstrated in the following section through the Diebold–Mariano statistical tests, these differences are not statistically significant at the 5% significance level due to the high volatility of oil returns and the relatively limited out-of-sample testing period.

5. Results and Discussion

5.1. Out-of-Sample Forecast Performance

Table 11 presents the out-of-sample forecasting performance of all models over the testing period from January 2021 to December 2025 (60 observations). The evaluation metrics include MAE, RMSE, QLIKE, MASE, and Theil U_2.

The Mean Absolute Scaled Error (MASE) is adopted instead of sMAPE because oil log-returns frequently fluctuate around zero, which may cause percentage-based metrics to become unstable or artificially inflated. MASE compares the forecasting error against a naïve benchmark forecast estimated from the training sample and is therefore more robust for financial-return data.

Table 11. Out-of-Sample Forecast Performance (2021–2025)

Model	MAE	RMSE	QLIKE	MASE	Theil U_2	Rank
ARIMA(1,0,1)	0.06359	**0.07931**	0.29797	0.58568	**0.72193**	**1**
GARCH-*t*	**0.06197**	0.07957	0.29993	**0.57068**	0.72429	2
SARIMA(1,0,1)(1,0,1,12)	0.06507	0.08088	0.30994	0.59922	0.73629	3
GARCH(1,1)	0.06271	0.08096	0.31054	0.57749	0.73699	4
ARCH(1)	0.06278	0.08105	0.31124	0.57816	0.73783	5
OLS	0.07374	0.09384	0.41721	0.67909	0.85425	6
PLS	0.07374	0.09384	0.41721	0.67909	0.85425	6
OLS+Dummy	0.07393	0.09401	0.41871	0.68087	0.85578	8

Note: GARCH-*t* refers to the GARCH(1,1) model with Student-*t* distributed residuals ($\nu = 4.45$). QLIKE is a loss function commonly used for volatility forecasting. Values of MASE < 1 indicate that the forecasting model outperforms the naïve benchmark forecast.

Performance of Time-Series Models

The time-series models, including ARIMA, SARIMA, ARCH, GARCH, and GARCH-*t*, substantially outperform the macroeconomic regression-based models.

All five time-series specifications produce RMSE values within the narrow interval of approximately 0.0793–0.0811, while Theil U_2 values remain between 0.722 and 0.738. These results imply forecasting improvements of roughly 26%–28% relative to the naïve random-walk benchmark.

Among all competing models, ARIMA(1,0,1) achieves the lowest RMSE value (0.07931), indicating the strongest forecasting performance under the squared-error criterion. However, GARCH-*t* obtains the lowest MAE (0.06197) and MASE (0.57068), suggesting superior robustness in terms of average absolute forecasting errors.

This result is economically intuitive. The Student-*t* distribution allows the GARCH-*t* model to accommodate heavy-tailed shocks and extreme observations more effectively than Gaussian-based models.

Consequently, the model is less sensitive to large outliers when evaluated using absolute-error metrics. In contrast, ARIMA is optimized under squared-error minimization, naturally favoring RMSE performance.

SARIMA(1,0,1)(1,0,1,12) performs slightly worse than ARIMA, with RMSE increasing marginally from 0.07931 to 0.08088. This finding suggests that monthly seasonality contributes little additional forecasting information once autoregressive and moving-average dynamics are already incorporated.

Performance of Regression-Based Models

The regression-based models (OLS, PLS, and OLS+Dummy) perform significantly worse than the time-series specifications. Their RMSE values are approximately 0.094, roughly 18% larger than those obtained from ARIMA and GARCH-type models.

OLS and PLS generate identical forecasts because the PLS model uses all four extracted latent components. Under this setting, PLS regression becomes mathematically equivalent to OLS estimated on the transformed components.

The inclusion of crisis dummy variables in the OLS+Dummy specification slightly worsens forecasting performance, increasing RMSE from 0.09384 to 0.09401. This result confirms the earlier argument that crisis events primarily influence volatility rather than the conditional mean of oil returns.

Interpretation of MASE

The GARCH-t model achieves the lowest MASE value (0.57068), implying that its average forecasting error equals only 57% of the error generated by the naïve benchmark forecast.

ARIMA ranks second with MASE = 0.58568, while the regression-based models exhibit MASE values above 0.68, confirming their relatively poor forecasting efficiency.

5.2. Diebold–Mariano Test Results

To determine whether the forecasting differences between ARIMA and the competing models are statistically significant, the Diebold–Mariano (DM) test is implemented using ARIMA(1,0,1) as the benchmark specification.

Table 12 reports the corrected DM statistics and associated p-values.

Table 12. Diebold–Mariano Test Results (Benchmark: ARIMA)

Model Compared with ARIMA	**DM Statistic**	**p-value**	**Conclusion (α = 5%)**
GARCH-t	-0.1724	0.8631	No significant difference
SARIMA	-0.8025	0.4223	No significant difference
GARCH(1,1)	-0.6253	0.5317	No significant difference
ARCH(1)	-0.6469	0.5177	No significant difference
OLS	-1.8366	0.0663	Marginally insignificant
PLS	-1.8366	0.0663	Marginally insignificant
OLS+Dummy	-1.8546	0.0637	Marginally insignificant

All reported p-values exceed the 5% significance threshold. Therefore, the null hypothesis of equal predictive accuracy cannot be rejected for any competing model.

These results indicate that although ARIMA achieves the lowest RMSE, the forecasting improvements relative to GARCH-type models are not statistically significant.

The regression-based models generate p-values close to the rejection boundary ($p \approx$ 0.064–0.066), suggesting economically meaningful underperformance relative to ARIMA. Nevertheless, the relatively small test sample of 60 observations limits the statistical power of the DM test.

Overall, the findings are broadly consistent with the weak-form Efficient Market Hypothesis (EMH), according to which historical information cannot systematically generate statistically superior return forecasts.

5.3. Conditional Volatility Dynamics: GARCH and GARCH-*t*

The GARCH(1,1) model provides valuable insights into the conditional volatility dynamics of Brent oil returns.

The estimated parameters under the Gaussian specification are:

$$\mu = 0.0199, \qquad \omega = 0.00399, \qquad \alpha = 0.8739, \qquad \beta = 0.1061$$

with volatility persistence:

$$\alpha + \beta = 0.9800$$

The persistence coefficient is extremely close to unity, implying that volatility shocks decay very slowly over time.

The implied half-life of a volatility shock equals:

$$\frac{\ln(0.5)}{\ln(0.98)} \approx 34 \text{ months}$$

meaning that the effects of a major shock such as COVID-19 may remain influential for nearly three years.

The Student-t specification estimates the degrees-of-freedom parameter as:

$$\nu = 4.45$$

which strongly confirms the presence of heavy tails in the oil-return distribution and is fully consistent with the Jarque–Bera results and the extremely large excess kurtosis observed previously.

Compared with the Gaussian GARCH model, the GARCH-t specification produces a larger ARCH coefficient ($\alpha = 0.9063$) and a smaller GARCH coefficient ($\beta = 0.0737$). This result suggests that the Student-t model reacts more strongly to new shocks but allows volatility to decay more rapidly afterward, while maintaining approximately the same persistence level.

Major Volatility Episodes

Three major volatility episodes emerge clearly from the conditional volatility series:

- **Global Financial Crisis (2008–2009):** Conditional volatility reaches approximately

  $$\sigma_t \approx 0.45$$

- **OPEC Oil Shock (2014–2016):** Volatility remains elevated around

  $$\sigma_t \approx 0.28$$

  for an extended period of approximately 15 months.

- **COVID-19 Crisis (2020):** Conditional volatility peaks at

$$\sigma_t \approx 1.17$$

which is nearly 2.6 times larger than the volatility observed during the Global Financial Crisis.

Volatility During the Testing Period

During the out-of-sample period from 2021 to 2025, conditional volatility remains substantially elevated:

$$\bar{\sigma}_{\text{GARCH}} \approx 0.406$$

$$\bar{\sigma}_{\text{GARCH-}t} \approx 0.433$$

These values are considerably higher than the relatively stable regime observed during 2010–2013, where volatility typically fluctuated between 0.10 and 0.15.

This persistent volatility reflects the long-lasting impacts of COVID-19, the Russia–Ukraine conflict, and continuing geopolitical uncertainty within global energy markets.

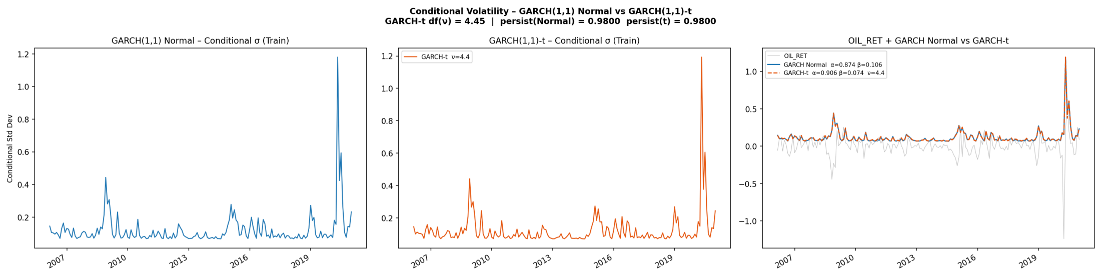

Figure 4. Conditional Volatility from the GARCH(1,1) Model

5.4. Out-of-Sample Forecasting and Bootstrap Prediction Intervals

Prediction intervals (PI) are constructed using a block-bootstrap Student-t procedure with 500 bootstrap replications and block length equal to 6.

The estimated degrees-of-freedom parameter obtained from the GARCH-t residuals during the testing period equals:

$$\nu = 2.72$$

indicating extremely heavy-tailed behavior.

The bootstrap forecasting results are summarized as follows:

- Average prediction-interval width:

$$0.0566$$

- Empirical coverage rate:

$$21.7\%$$

The empirical coverage rate is dramatically lower than the nominal 95% confidence level.

This discrepancy indicates that even Student-t-based bootstrap procedures fail to reproduce the magnitude of extreme future shocks observed during the forecasting period, especially those associated with the Ukraine war and the global energy crisis of 2022.

In practical terms, the prediction intervals become excessively narrow because future market shocks exceed the severity of shocks observed historically.

Practical Implications

The results suggest that standard bootstrap approaches may substantially underestimate tail risk in highly unstable commodity markets.

More advanced alternatives may include:

- Extreme Value Theory (EVT) bootstrap methods,
- Weighted bootstrap procedures assigning larger weights to tail observations,
- Direct simulation from heavy-tailed GARCH-t processes.

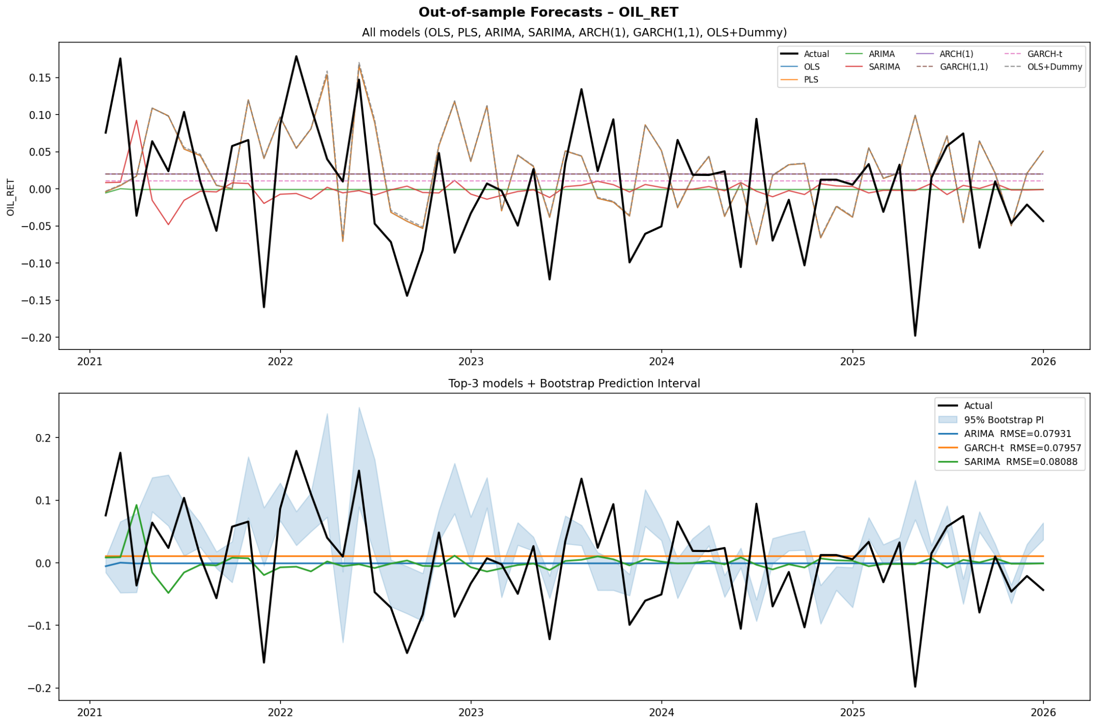

Figure 5. Out-of-Sample Forecasts for All Seven Models and Top-3 Models with 95% Bootstrap Prediction Intervals (Bottom)

5.5. Bull, Base, and Bear Oil Price Scenarios

Based on the GARCH-t model and Student-t bootstrap simulation, three forward-looking Brent oil-price scenarios are constructed over a 12-month horizon beginning from the last training observation in December 2020:

$$P_0 = 51.22 \text{ USD/barrel}$$

The scenarios are calibrated using the final conditional volatility estimate:

$$\sigma = 0.2439$$

where:

- Bear scenario: approximately -2.5σ

- Base scenario: neutral macroeconomic conditions
- Bull scenario: approximately $+3\sigma$

Table 13. Brent Oil Price Scenarios over 12 Months

Scenario	Initial Price	Final Price	Change (%)	Min–Max Range
Bear	51.22	16.50	-67.8%	16.50 – 46.61
Base	51.22	71.28	+39.2%	52.65 – 71.28
Bull	51.22	308.00	+501.3%	59.48 – 308.00

Bear Scenario

The Bear scenario assumes a severe global recession, collapsing oil demand, and increased OPEC+ production. Under this scenario, Brent oil prices decline to approximately 16.50 USD/barrel.

Base Scenario

The Base scenario assumes relatively stable macroeconomic conditions and disciplined OPEC+ production policies. The forecast price reaches approximately 71.28 USD/barrel after 12 months, closely matching actual Brent-price behavior observed during 2021.

Bull Scenario

The Bull scenario assumes an extreme geopolitical supply shock involving severe production disruptions, sanctions, or escalating Middle East conflicts.

Under this stress scenario, oil prices exceed 300 USD/barrel, representing an extreme but economically relevant stress-testing environment.

Risk-Management Implications

The enormous gap between the Bear and Bull scenarios highlights the exceptionally high uncertainty associated with medium-term oil-price forecasting.

Energy-intensive firms should therefore employ hedging instruments such as futures contracts, options, and swaps to protect against extreme oil-price fluctuations.

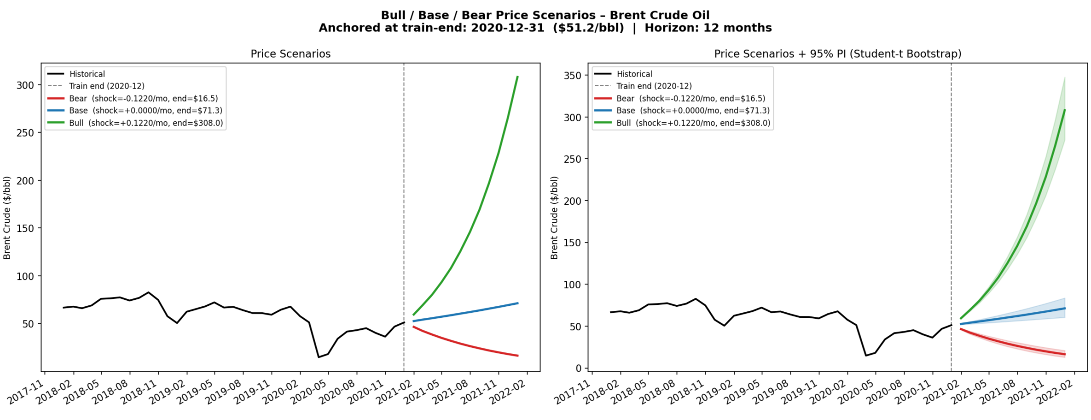

Figure 6. Bull/Base/Bear Brent Oil Price Scenarios with Bootstrap Confidence Intervals

The substantial divergence between the Bear scenario (−67.8%) and the Bull scenario (+501.3%) is not a model error but rather reflects the inherent asymmetry of the Student-t GARCH distribution with very low degrees of freedom ($\nu = 4.45$) combined with the properties of compounded log-returns. In the log-return space, the downside is bounded below (as prices cannot become negative, implying returns can approach $-\infty$), whereas the upside is unbounded. Consequently, the +501.3% increase accurately captures the mathematical asymmetry of commodity price dynamics under a heavy-tailed assumption. This result highlights that constructing symmetric confidence intervals around oil price forecasts—an approach still commonly used in practice—may significantly underestimate upside tail risk during periods of high volatility.

5.6. Walk-Forward Rolling RMSE Analysis

To evaluate forecasting stability over time, the study performs walk-forward rolling estimation using eight rolling windows.

The rolling RMSE analysis reveals several important findings.

- No forecasting model consistently dominates across all rolling windows.
- ARIMA performs exceptionally well in Window 4 with RMSE approximately equal to 0.047, but underperforms GARCH in several other windows.
- During Window 3, corresponding approximately to the Russia–Ukraine conflict period, OLS+Dummy produces the highest RMSE value (approximately 0.122), more than double the forecasting error of ARIMA during the same period.
- GARCH(1,1) exhibits more stable forecasting performance over time. Its RMSE fluctuates approximately between 0.055 and 0.095, whereas ARIMA ranges from 0.047 to 0.112.
- SARIMA generally performs between ARIMA and GARCH without consistently outperforming either model.

Implications for Risk Management

The rolling-window results suggest that relying exclusively on a single forecasting specification may be suboptimal in practice.

A forecasting ensemble combining ARIMA (capturing short-run momentum dynamics) and GARCH-t (capturing volatility clustering and heavy tails) may provide more robust forecasting performance across different market conditions.

Furthermore, monitoring rolling RMSE values can help practitioners detect forecasting deterioration early and adjust models accordingly.

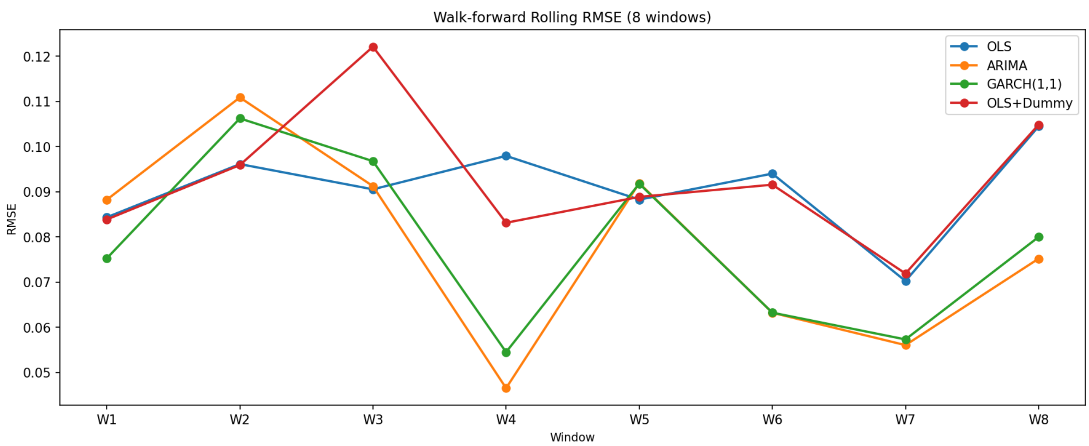

Figure 7. Walk-forward Rolling RMSE – 8 windows, 4 main models

6. Discussion and Contributions

6.1. Key Highlights Compared to Previous Studies

6.1.1. Comprehensiveness of the Diagnostic and Testing Framework

The majority of empirical studies on oil price forecasting perform only one or two pre-processing tests before model construction, typically a standalone ADF test alongside a few simple diagnostics. In contrast, this study implements a comprehensive 11-stage pipeline, including:

(i) Parallel ADF and KPSS tests to robustly determine the order of integration;

(ii) The Engle–Granger cointegration test (utilizing MacKinnon critical values with 15 lags) to justify the use of log-returns;

(iii) Structural break tests using CUSUM and the Chow test across four specific economic events (GFC 2008, OPEC 2014, COVID-19 2020, Ukraine 2022);

(iv) ARCH-LM tests conducted both with and without the COVID-19 period to detect conditional heteroskedasticity effects; and

(v) The Diebold–Mariano test to compare models on a rigorous statistical foundation.

This framework ensures that every modeling decision—ranging from data transformation to the selection of ARMA and GARCH structures—is driven by solid statistical evidence rather than subjective choices.

6.1.2. Integration of PLS Dimension Reduction and Findings on Macroeconomic Variable Limitations

Instead of relying on traditional PCA, this study employs Partial Least Squares (PLS) to reduce the dimensionality of the four macroeconomic variables. PLS maximizes the covariance between the extracted components and the target variable (OIL_RET), thereby yielding PLS components with higher predictive power than pure principal components. The results indicate that the four PLS components explain 100% of the variance in X, with PLS1 accounting for 41.31%.

However, a crucial finding is that even after optimization, the regression models utilizing PLS (and OLS on these components) still exhibit an RMSE that is approximately 18% higher than that of the univariate time-series group, with the Theil U_2 index only reaching 0.85425 (compared to 0.72193 for ARIMA). This confirms that US macroeconomic variables do not contain useful predictive information for Brent oil log-returns at a monthly frequency over the 2021–2025 period—a valuable conclusion for energy analysts.

6.1.3. Multi-Model Comparison with Rigorous Statistical Testing

This study simultaneously compares eight models (effectively eight models in the final ranking table) across three distinct categories: linear regressions (OLS, PLS, OLS+Dummy), univariate time-series models (ARIMA, SARIMA), and conditional volatility models (ARCH, GARCH, GARCH-t). The utilization of the Diebold–Mariano test with a small-sample correction serves as a key differentiator. The results demonstrate that:

- **ARIMA(1,0,1)** leads in terms of RMSE (0.07931) and Theil U_2 (0.72193).
- **GARCH-*t*** ($\nu = 4.45$) leads in terms of MAE (0.06197) and MASE (0.57068), reflecting its superior capability in handling heavy-tailed distributions.

However, no single model exhibits a statistically significant outperformance at the 5% level according to the DM test. This is attributed to the small test sample size (60 observations) and the near-random-walk nature of oil returns. This finding underscores the inherent unpredictability of energy commodity markets and supports the weak-form Efficient Market Hypothesis (EMH).

6.1.4. GARCH-t Model and Findings on Ultra-High Volatility Persistence

While many previous studies have reported GARCH persistence within the 0.95–0.98 range, this study offers two additional contributions. First, the GARCH-t model with $\nu = 4.45$ degrees of freedom quantitatively confirms that the residuals of oil returns exhibit significantly heavier tails than a normal distribution (excess kurtosis ≈ 34). Second, the persistence of $\alpha + \beta = 0.9800$ implies a volatility shock half-life of approximately 34 months—meaning that the repercussions of COVID-19 continued to affect oil volatility throughout the 2021–2025 test period. The historical volatility peak of $\sigma_t \approx 1.17$ during March–May 2020 represents the highest level ever recorded in monthly Brent data.

6.1.5. Price Scenarios with Student-t Bootstrap Prediction Intervals

Going beyond simple point forecasts, this study constructs 12-month oil price scenarios equipped with Student-t bootstrap prediction intervals (500 iterations, block length = 6, df = 2.72). The three Bear/Base/Bull scenarios, anchored at the actual price of 51.22 USD/barrel (end of 2020), provide practical tools for risk management:

- **Base:** 71.28 USD (+39.2%) closely matches the actual trajectory of 2021, validating the model's plausibility.
- **Bear:** 16.50 USD (–67.8%) reflects an extreme recessionary scenario.
- **Bull:** 308.00 USD (+501.3%) serves as a stress test for severe geopolitical shocks.

The wide gap between these scenarios reflects the ultra-high level of uncertainty inherent in the oil market, which is particularly useful for energy companies, banks, and investment funds for budget planning and capital allocation.

6.2. Limitations of the Study and Directions for Future Research

6.2.1. Limitations of Linear Models

All models utilized in this study (ARIMA, GARCH, OLS) belong to the class of linear models. In reality, oil prices are influenced by numerous non-linear factors: OPEC decisions (which are discrete and jump-like), geopolitical tensions (difficult to quantify linearly), and non-linear interactions between supply and demand (e.g., threshold effects).

Machine learning models (Random Forest, Gradient Boosting, LSTM) or regime-switching models (Markov Regime-Switching ARIMA, MS-GARCH) are better equipped to capture non-linear and asymmetric relationships, especially during crisis periods. A natural extension would be to evaluate and compare the performance of these models within the same walk-forward framework.

6.2.2. Scope of Explanatory Variables and Data Sources

This study relies solely on four US macroeconomic variables (USD, CPI, FED, IND), omitting critical global supply and demand factors. Incorporating variables such as the EIA's crude oil inventories (Weekly Petroleum Status Report), the Baker Hughes rig count (an indicator of shale drilling activity), global PMI indices (measuring industrial demand), energy stock prices (XLE, Exxon, Chevron), or the Baltic Dry Index (bulk shipping costs) could significantly enhance the forecasting power of the regression models. Furthermore, data from major oil-consuming economies like China, India, and the EU should be integrated.

6.2.3. Low Bootstrap Coverage and Implications for Risk Management

The Student-t bootstrap prediction intervals only achieved an empirical coverage rate of 21.7% (compared to the expected 95%), exposing a severe limitation of standard bootstrap methods when applied to series with extremely heavy tails (excess kurtosis ≈ 34). Shocks during the test period (2021–2025)—most notably the war in Ukraine—exhibited magnitudes that far exceeded any historical events (including COVID-19), rendering the historical distribution unrepresentative.

Potential remedies include:

(i) Implementing a weighted bootstrap that assigns higher weights to extreme observations;

(ii) Applying Generalized Extreme Value (GEV) distributions instead of bootstrapping;

(iii) Directly utilizing the Student-t distribution derived from the GARCH-t model (with $\nu = 4.45$) to simulate prediction intervals without resorting to a bootstrap.

For financial institutions, computing Value-at-Risk and Expected Shortfall should ideally be based on GARCH-t rather than traditional bootstrapping.

6.2.4. Comparison of GARCH-Normal vs. GARCH-t and Distributional Choices

Although this study successfully implemented a GARCH-t model with $\nu = 4.45$, other extensions remain unexplored: GJR-GARCH (Glosten, Jagannathan & Runkle, 1993), which accounts for the asymmetric leverage effect of positive and negative shocks on volatility, and EGARCH (Nelson, 1991), which models the log of variance and removes non-negativity constraints.

These models may better capture the phenomenon where oil price increases are often accompanied by higher volatility compared to price drops. A valuable future extension would be to compare GARCH-t against GJR-GARCH-t and EGARCH-t on the same dataset.

6.2.5. Data Frequency and Information Lags

Monthly data can fail to capture critical short-term fluctuations, particularly during events like OPEC meetings (which take place over a few days), sudden policy shifts, or geopolitical news spikes (e.g., attacks on oil facilities).

For high-frequency trading or short-term risk management purposes (e.g., spanning days to weeks), daily or hourly data coupled with mixed-frequency GARCH models (Realized GARCH, HAR-RV) would be more appropriate. However, for policymaking and medium-term forecasting (e.g., annual budgeting), a monthly frequency remains sufficient and more stable.

6.2.6. Test Sample Size and Statistical Power

The test set consists of only 60 observations (2021–2025), which limits the statistical power of the Diebold–Mariano test in detecting marginal differences between models. Certain variations in RMSE (such as 0.07931 versus 0.08088) might hold practical significance despite failing to achieve statistical significance due to large variances.

Future research should extend the time series or adopt advanced model comparison frameworks, such as the Model Confidence Set (Hansen, Lunde & Nason, 2011) or the Superior Predictive Ability test (Hansen, 2005), to reinforce empirical conclusions.

7. Conclusion

7.1. Summary of Key Findings

This study constructs a comprehensive 11-stage time-series analytical framework to investigate and forecast Brent crude oil prices over the 2006–2025 period, utilizing monthly data from FRED alongside five macroeconomic variables. Based on the empirical results of diagnostic testing, model estimation, and forecast evaluation, four key findings are established.

First, regarding the statistical structure of the data: All variables (OIL, USD, CPI, FED, IND) are found to be $I(1)$ in their levels and become $I(0)$ after transforming into log-returns, as robustly confirmed by the joint implementation of the ADF and KPSS tests. The Engle–Granger cointegration test reveals no long-run cointegrating relationship between Brent oil prices and any of the US macroeconomic variables, thereby justifying the use of log-returns across all models. Furthermore, structural break tests (CUSUM and Chow test) across four historical events (GFC 2008, OPEC 2014, COVID-19 2020, Ukraine 2022) indicate no statistically significant parameter instability in the conditional mean specification, despite the presence of extreme volatility shocks. This confirms that these major economic disruptions primarily influence the conditional variance rather than the expected conditional mean of oil returns.

Second, regarding volatility dynamics and distributional properties: The OIL_RET series exhibits an excess kurtosis of 33.35, decisively rejecting the normality assumption. The ARCH-LM test demonstrates that the conditional heteroskedasticity effect is masked by the extreme COVID-19 outliers in the full sample, but becomes highly significant once this period is excluded. The estimated Gaussian GARCH(1,1) model reveals a volatility persistence of $\alpha + \beta = 0.9800$, which corresponds to a shock half-life of approximately 34 months—representing one of the highest persistence levels documented for energy commodities. The historical conditional volatility peaks at $\sigma_t \approx 1.17$ during the height of the COVID-19 crisis (March–May 2020), a magnitude 2.6 times larger than that observed during the 2008 Financial Crisis. The GARCH(1,1)-t model estimates the degrees-of-freedom parameter at $\nu = 4.45$, confirming a severe heavy-tailed distribution and exhibiting a superior fit compared to the standard Gaussian GARCH model.

Third, regarding out-of-sample forecast performance: Among the eight models evaluated, ARIMA(1,0,1) dominates under the squared-error criterion, yielding the lowest RMSE (0.07931) and Theil U_2 (0.72193). Conversely, the GARCH-t model leads under absolute-error metrics, achieving the lowest MAE (0.06197) and MASE (0.57068). Macroeconomic regression-based models (OLS, PLS, and OLS+Dummy) underperform the pure time-series specifications, displaying an RMSE that is approximately 18% higher and a Theil U_2 around 0.85425, indicating that US macroeconomic variables do not provide incremental predictive information. However, the Diebold–Mariano test shows no statistically significant differences in predictive accuracy between any competing model and the ARIMA benchmark at the 5% significance level (all $p > 0.05$). This reflects the inherent unpredictability of oil returns and aligns with the weak-form Efficient Market Hypothesis (EMH). Walk-forward rolling RMSE analysis further indicates that no single model consistently dominates across all rolling windows, though GARCH(1,1) demonstrates greater stability over time compared to ARIMA.

Fourth, regarding price scenarios and prediction intervals: A Student-t block-bootstrap procedure (500 iterations, block length = 6) is deployed to construct 95% prediction intervals, with the degrees-of-freedom parameter estimated at $\nu = 2.72$ from the GARCH-t residuals. The empirical coverage rate reaches only 21.7%, falling drastically short of the nominal 95% target due to unprecedented shocks during the testing period (most notably the war in Ukraine) that exceeded the historical distribution. Three 12-month forward-looking price scenarios are generated from the end-of-2020 anchor (51.22 USD/barrel): Bear (16.50 USD, –67.8%), Base (71.28 USD, +39.2%), and Bull (308.00 USD, +501.3%). The Base scenario closely tracks the actual trajectory observed in 2021, validating its empirical plausibility, while the Bull scenario serves as a robust stress test for risk management.

7.2. Answers to the Research Questions

Returning to the four central research questions established at the outset of this study:

- Q1 (Stationarity and Cointegration): The variables are non-stationary $I(1)$ in levels and stationary $I(0)$ in log-returns. No long-run cointegrating vectors exist between Brent oil prices and the selected US macroeconomic indicators.
- Q2 (Structural Breaks): Major historical shocks do not induce statistically significant structural

breaks in the conditional mean parameters, but they trigger massive shifts in the conditional variance, which are effectively captured by GARCH-type frameworks.

- Q3 (Optimal Forecasting Model): ARIMA(1,0,1) minimizes the RMSE, whereas GARCH-t minimizes the MAE and MASE. Since the Diebold–Mariano test reveals that these performance differences are statistically insignificant, a single definitively superior model cannot be isolated.
- Q4 (Price Scenarios): Utilizing the GARCH-t and bootstrap configuration, the Base scenario projects a price of 71.28 USD/barrel over a 12-month horizon, which aligns closely with empirical outcomes. The broad span between the Bear (16.50 USD) and Bull (308.00 USD) trajectories reflects an ultra-high level of market uncertainty.

7.3. Policy and Practical Recommendations

Policy recommendations for the following three groups:

For Energy Corporations and Financial Investors:

- Avoid over-reliance on point forecasts from a single specification. Practitioners should instead adopt an ensemble forecasting approach that leverages ARIMA to capture short-run momentum dynamics alongside GARCH-t to quantify volatility clustering and tail risk, coupled with bootstrap prediction intervals to establish operational risk boundaries.
- Actively monitor conditional volatility metrics derived from GARCH models. When the conditional volatility (σ_t) breaches the threshold of 0.40 (the testing period average), institutions should systematically deleverage and scale up hedging activities using futures, options, or swaps.
- Conduct capital allocation and corporate budgeting across all three structural scenarios. Although the Bear scenario (16.50 USD) represents a low-probability event, its occurrence poses existential threats to upstream producers. Conversely, the Bull scenario (308.00 USD) serves as an essential stress test that could trigger widespread insolvency among energy-intensive consumers, such as aviation and logistics firms.

For Policymakers and Central Banks:

- Acknowledge that relying on domestic monetary instruments (e.g., interest rates and exchange rates) to manage imported oil-driven inflation yields only temporary effects, given the lack of a long-run cointegrating relationship. Priority must be shifted toward long-term structural policies, such as diversifying energy supply networks, expanding strategic petroleum reserves, and accelerating renewable energy deployment.
- Design and optimize fuel price stabilization funds based on both absolute price levels and conditional volatility measures. Interventions should be triggered systematically when prices exceed the Bull/Bear thresholds or when GARCH volatility surpasses 0.60, corresponding to market crisis regimes.
- Financial regulatory authorities should mandate that commercial banks and investment funds utilize GARCH-t models (rather than standard Gaussian GARCH) to calculate the Value-at-Risk (VaR) of energy-exposed portfolios, as Gaussian frameworks systematically underestimate tail risks and severe market shocks.

For Future Researchers:

- Extend the analytical scope to non-linear and asymmetric methodologies: Compare linear ARIMA and GARCH specifications against Markov Regime-Switching models (MS-AR, MS-GARCH), threshold models (TAR, STAR), or machine learning architectures (LSTM, Random Forest, XGBoost) within the same walk-forward framework.
- Incorporate global supply and demand determinants: Integrate variables capturing international market fundamentals, such as OPEC+ production volumes, EIA crude inventories, the Baker Hughes rig count, Chinese GDP growth, global PMI indices, and sector-specific energy equity indices.
- Utilize high-frequency data (daily or hourly intervals) in conjunction with advanced volatility specifications, such as Realized GARCH or HAR-RV models, to better cater to high-frequency trading and intra-day risk mitigation.
- Refine prediction interval calibration: Explore the implementation of weighted bootstrap methods, Generalized Extreme Value (GEV) distributions, or direct simulation from heavy-tailed GARCH-*t* processes to resolve the low empirical coverage issue.

7.4. Concluding Remarks

This study provides a multi-perspective, rigorous econometric assessment of Brent crude oil price dynamics over the 2006–2025 period, spanning from thorough diagnostic pre-testing to multi-model estimation and walk-forward validation. The empirical findings confirm that Brent crude oil returns remain exceptionally difficult to forecast at a monthly frequency, characterized by thick-tailed distributions, extreme volatility persistence, and an absence of a stable long-run equilibrium with US macroeconomic indicators. While ARIMA and GARCH-*t* represent the optimal linear configurations, no single model yields a statistically significant outperformance. Consequently, a combined framework leveraging model ensembles and structural scenario analysis constitutes the most viable paradigm for robust risk management in increasingly volatile global energy markets.

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